Available online at www.sciencedirect.com
ScienceDirect
Mathematics and Computers in Simulation 185 (2021) 1–16
MATHEMATICS AND COMPUTERS IN SIMULATION
www.elsevier.com/locate/matcom

Original articles

FPGA based effective agriculture productivity prediction system using fuzzy support vector machine

G. Prabakaran[a,*], D. Vaithiyanathan[b], Madhavi Ganesan[c]

[a] Department of Electronics and Communication Engineering, College of Engineering, Guindy, Anna University, Chennai 600025, India
[b] Department of Electronics and Communication Engineering, National Institute of Technology Delhi, New Delhi 110040, India
[c] Centre for Water Resources, College of Engineering, Guindy, Anna University, Chennai 600025, India

Received 1 April 2020; received in revised form 10 December 2020; accepted 12 December 2020
Available online 19 December 2020

Abstract

This work investigates the functions of hardware-implemented intelligent decision support system using support vector machines. The system aims to forecast future productivity based on the data prepared by field experts followed by productivity influence factors. This feature is perceived by the combination of fuzzy logic and support vector machine. The proposed approach has been thoroughly tested at a ground level, and the designed structural test results have made major improvements compared to the lack of proper approach. This system proposed to compensate for performance decrease, achieved higher productivity with a prediction accuracy of 95%. Furthermore, the proposed intelligent embedded decision support system provided the deficit level of needed input scale to increase productivity and avoid excess consumption of fertilizer in agriculture. A 30-year climate parameter has been taken into account to establish such a system to control the consumption of fertilizers.
© 2020 International Association for Mathematics and Computers in Simulation (IMACS). Published by Elsevier B.V. All rights reserved.

Keywords: Support vector machine; FPGA; Fuzzy logic; Intelligent system; Decision support system (DSS); Hardware implementation

1. Introduction

Information and technological development are among the best in providing many new facilities and enthusiastic opportunities for humanity and society. Over the past few years, farmers, agriculture experts, and agriculture researchers have been investigating the use of advanced information technologies to improve agriculture [25,37]. Owing to the use of minimum level of input energy, increasing the maximum productivity process is more complicated. However, it has happened, productivity can become sometimes distressed even if it takes a higher level of failure and continues to operate in that condition for many seasons. In particular agricultural lands meet or have faced difficulty in producing the quantum of output due to an imbalance of nutrient pattern [53]. Therefore nutrient imbalance from time to time can lead to crop failure. Crop failure occurs immediately after a high distressed climatic condition. Several factors have influenced crop growth potential. They are nutrient imbalance, climatic conditions, pest incidence, seasonal factors, soil and water factors. Overall, it affects the whole thing, and the way it affects is measured by the quality or quantity variables they have attained. As a result, the system accurately predicts whether the crop productivity achieves higher quantum or not. If the predicted results are minimum productivity then the

* Corresponding author.
E-mail address: kgp.com@outlook.com (G. Prabakaran).

https://doi.org/10.1016/j.matcom.2020.12.011
0378-4754/© 2020 International Association for Mathematics and Computers in Simulation (IMACS). Published by Elsevier B.V. All rights reserved.

designed system will make recommendations for achieving maximum productivity or recommend alternative crops. At the same time, agricultural productivity measurement event is a complex process it was derived from various literary criticisms and knowledge. Therefore, there are a lot of difficulties in obtaining high production in agriculture. Despite the problems, it is necessary to adjust, and this seems to be possible, and many of these technologies have been used and the right output is derived.

Agriculture does not only depend on a single factor; instead, it consists of many of factors such as soil, climate, fertilizer, water, etc. Therefore, in real-world phenomena, agricultural problems are complicated by uncertainty. The probability theory is applied and the uncertainty conditions are handled. But it is only used for situations that run based on random actions. In such a case, the fuzzy logic approach is the only method to handle similar instances like agriculture. But at the same time, it cannot be used directly because it has separate classes with different categories followed by various problems. Hence, soft computing techniques and their constituents like support vector machines, artificial neural networks, Naive Bayes and k-nearest neighbor classifier methods are used to achieve their merit output [24,29,45,56]. At the same time, it is not possible to design a separate system with a mechanism to solve each problem. On the other hand, it is necessary to bring together all the agriculture parameters into one system. This paper introduces effective agriculture predicted system for improving agricultural productivity and adopts the fuzzy support vector machine approach with the support of real-time field values. The fuzzy combined SVM has already provided valid output in many agricultural applications such as drought prediction, weather warning system, crop price forecasting and irrigation water quality model [18,20,55,57]. Therefore, the proposed system includes crop prediction rule with crop influencing factors, recommendations, fuzzy logic and a support vector machine.

Planting in agriculture is the most appropriate and most important process. It includes the weather, climate, land, soil, nutrient, pest, but the inevitable is to plant the crop at the right time. Since all of these things are largely available in rural, therefore there is a tendency for them to do better. Accordingly, to improve the agriculture productivity, the unified system architecture is essential to make the farming more easier. Its dimensions have resulted in better output, namely categories like a decision support system (DSS), Information System Simulation (ISS) and site-specific system [16,31]. Therefore the intellectual information system in the form of DSS continues to play an essential role in the development of agricultural products, management, and diseases diagnosis and yield prediction [6,21,30,39]. Hence, the purpose of this study is to enhance the production of different crops in agriculture, develop farming knowledge, improve embedded intelligence decision making and forecasting, and utilize an accurate predictive model at least for the farmland benefits cost ratio like anything else. Generally, the term BC ratio is related to the concept of economic conditions. In particular, it is directly involved in the market prices and inputs required to grow the respective crop season after a certain period and thereby excluding the cost of producing the crop. Therefore, the benefit–cost ratio is calculated by using the standardized formula as in [46]. Thus, the main objective of this proposed system is to calculate the extent of production before the cropping in all kinds of conditions and other nutrients that are useful for the growth of crops. Besides, this research has been fall with all the background of the following objectives.

1. Optimum fertilizer usage and increase agricultural productivity.
2. An FSVM based embedded system to achieve a high benefit–cost ratio (BCR).
3. Support Vector Machine-based fuzzy system to predict farm productivity using FPGA.

This work also attempts to explain the problem of such an extended period in long-term surveillance to prevent a series of failures through a systematic approach. Initially, two agro-climatic zones (ACZ) are selected for the determination of the site experiment. The selected ACZ's is further divided into sub-regions, where drip irrigation has been set up and the soil fertility and water quality are inspected, and strategies are framed for crop cultivation. As per the aforesaid information, SVM will overcome the ambiguity and develop the crop influenced factors along with the fuzzy logic. Therefore, the intelligent embedded decision support system addressed the future throughput of productivity for avoiding crop failure in the routine farming process. The system measures various parameters such as rainfall, humidity, temperature, soil properties, water properties, and pest incidence with fuzzy concepts as it approaches the future throughput transition. Hence, various parameters required for agricultural production are formed in such an intelligent system, measured and converted to a Register Transfer Level design (RTL). Also, this RTL has been synthesized and its hardware logic resource usage has been detected with a sequence of field-programmable gate arrays (FPGA). FPGA is an integrated circuit system designed to be built by a customer or

designer after production in the field. The role of the FPGA in the proposed system is to provide the valid decision-making in the stand-alone system even though if there are to handle multi-input and multi-output. The reconfigurable capacity of FPGA's has more added features for use of FPGA in the design of the system. The sequence of the programmable gates is as hefty as technology in the dimensional development of this electronic industry, and it has played a very positive role in forecasting. Therefore, the order of the programmable gate array has been used to convey the prediction in the proposed system. Likewise, the FPGA plays an important role in designing an intelligent system, such as a decision support system, because of the remarkable hardware structure and complexity of control devices. Simultaneously, the involvement of FPGA is numerous such as model prototype designs, automatic vehicle, and temperature measuring machines in line with industrial control equipment's [34].

Meanwhile, the support vector machine provides the learning statistics mechanism based on the learning theory. In addition, here, the support vector machine also provides the solid foundation for getting the IF-THEN rule, which depends on the rules generation method, satisfactory generalization capability and greater irreversible capability [17,54]. The fuzzy logic system is designed with multiple parameters so each of its parameters is approached in a comprehensive way in order to analysis the barriers to acceptance for system design [15]. On the other hand, electronic participation in agriculture is vital, and it helps to achieve various improvements in electronic tools, microprocessors, computer-based systems, and decision-making aids, including food, fodder, fiber, livestock, and post-harvest industry positions [4,40]. This results in the overall concentration of nutrients, which is considered the integrative successes of individual models.

In other similar conditions, the system of soft computing is in conjunction with the electronics industry that provides various technical applications. Besides, it is used in large quantities of agriculture applications such as surveillance system, crop growth assessment system, and site-specific treatment system [5,22]. The intuitive accuracy of FPGA is recommended by a fuzzy logic controller, which provides both systematic compatibility mode and accuracy in applications [15,32,35]. The recommendation of professional experts in the decision-support system is considered to be the most important because the decision-support system is based only on the recommendations of field experts [41,50]. The Xfuzzy environment is used to design the intelligent decision support system because this environment makes its way easier to specify the VHSIC hardware interpretation of the highest level from the beginning of the FIS specification, verification, package, and its primary notes [1,8,38,51]. The Xfuzzy is the platform consisting of tools to cover the several stages involved in the design process of fuzzy logic-based inference systems [13,52]. Thus, the undesirable assumption needed to be set up on the Xfuzzy platform, and the system's transfer status record is automatically generated [9]. Therefore, the primary purpose of the work is to manage various schemes and use the maximized available resources, which include multiple processes like digestion, digging and food [49]. In this study, a fuzzy controlled system has been built, which is designed with experienced database operators, along with these database models [26]. In respect to this, a small proportion of our three-year research decision is explained in a farmer-friendly initiative with technology penetration reported in [43]. In addition, it has already been designed as a fuzzy decision support system [44]. On one hand, the SVM offers solutions to many issues and the support of a soft computing system or decision support system [19,42].

Besides its precise predictive skill, it has been increasingly employed in technological applications, including predictions and fertilizer consumptions. If so, the SVM is used to combine the size of the product with fuzzy logic and the field-programmable gate array. Hence, the significance of the expression and the speed of the decision-making process, such as the Intelligent Embedded Decision Support system, are considered to be significant. Then it is necessary to increase the speed of decision in decision-making processes. The designed new system has shown the crop productivity of particular soil before the plantation of crop. Therefore, this proposed system addresses the development of a computational model using fuzzy support vector machine (FSVM) for improving agricultural productivity and this model has been implemented in embedded hardware. This leads to an awareness of nutrient deficiencies, soil problems, and adaptive climatic conditions. Furthermore, it also determines the nutritional deficiencies of soil and provides recommendations in a way to increase productivity.

1.1. Data collection

As per the government of Tamil Nadu Department of statistics, the following two agro-climatic zones have been chosen for this research work 1. Northeastern zone of Tamil Nadu and 2. The western zone of Tamil Nadu. Initially, the questionnaire has been designed to cultivate information about agriculture knowledge such as crop cultivation,

Table 1
Soil sample analysis report.

S.No	Parameter	Land 1	Land 2	Land 3	Land 4	Land 5	Land 6	Land 7	Land 8	Land 9	Land 10
1	pH	8.27	8.4	8.41	8.29	8.75	6.5	7.74	7.85	8.3	7.71
2	Electrical conductivity	0.18	0.45	0.301	0.146	0.266	0.285	0.359	0.09	0.181	0.085
3	Organic matter	0.66	1.11	0.99	1.12	0.72	0.56	0.98	0.89	1.55	0.365
4	Nitrate Nitrogen(ppm)	18.6	37.7	28.3	19.8	26.2	52.7	42.3	17.9	16.8	8.9
5	Available Phosphorus(ppm)	27.01	33.65	28.07	64.17	32.63	12.36	70.28	22.53	32.44	8.57
6	Potassium exchangeable(ppm)	197	181	155	129	227	187	307	89	281	56
7	Calcium exchangeable(ppm)	2135	2744	3048	2245	100	1042	1292	1136	2824	1257
8	Magnesium exchangeable(ppm)	938	1008	1028	772	337	359	441	440	500	394
9	Sulfur(ppm)	7.2	36.3	15.8	11.7	29.8	81.2	49.8	15.4	21.8	26
10	Sodium (ppm)	157	130	219	283	270	141	125	189	118	119
11	Zinc(ppm)	1.04	2.25	1.63	1.92	5.53	1.54	1.91	1.36	1.91	0.52
12	Manganese(ppm)	12.49	4.29	9.49	2.48	16.55	19386	10.12	8.96	5.9	18.79
13	Iron(ppm)	6.51	6.35	5.75	15.68	5.47	14.47	11.06	5.49	3.21	12.27
14	Copper(ppm)	2.18	2.72	2.34	2.59	1.83	2.72	1.82	1.53	2.45	1.93
15	Boron(ppm)	1	1.1	1.1	1.2	1.2	1	1	0.8	0.8	0.8
16	Molybdenum (ppm)	32.08	31.25	28.34	18.21	32.22	44.58	59.11	48.58	51.55	39.38

pest management, and season wise practices so on. To find the exact level of ability towards fertilizer knowledge, choice of the suitable crop for a favorable period and the ability of crop cultivation potential are evaluated through an open circuit channel interview with farmers. This survey is designed based on the previous research report, insights from professional experts and peasant recommendations. The peasants are selected based on the experience they have got, at least the minimum five decades of one particular crop cultivation. In that sense, some of the peasants have different opinions on applying the fertilizer. For instance, in paddy cultivation, peasant expert 1 suggests that even the category of primary nutrient moderate scale is provided sufficient to produce the maximum of productivity against the opinion of peasant expert 2. The reason behind the decision comes from soil organic matter concentration. As the peasant expert 2 strongly believes that soil organic matter concentration is quite crucial than other nutrient concerns. For the model design, 150 farmlands are selected based on our field experts' suggestions from January 2013 to December 2015. Accordingly, the 10 farmlands soil sample analysis report is provided in Table 1.

1.2. Data processing

After data collection, the fuzzy support vector machine approach is used for predicting agricultural productivity. Fuzzy support vector machine sorts the crop influenced parameters into predefined intervals. A total of 24 variables are used in the model building. The variables are selected because they had the importance of having created an impact on crop productivity as per the suggestions provided by the British Association meeting (1955) [2]. The 24 variables are divided into 8 subcategories. The variables are soil parameter variables, water parameter variables, nutrient parameter variables, climatic variables, pest incident parameter variables, and season parameter variables. The 25 variables are pH, Electrical conductivity, Organic matter, Nitrogen, Phosphorus, Potassium, Calcium, Magnesium, Sulfur, Sodium, Zinc, Manganese, Boron, Molybdenum, Copper, Rainfall, Temperature, Humidity, Pest, Rabi, Kharif, Summer, Soil and Water propriety variables, Productivity, and Recommendation. The relevant parameters of both the agro-climatic zone, including standard deviation and mean value, are shown in Table 2.

2. Fuzzy logic theory and fuzzy system

Fuzzy logic was introduced by Zadeh in 1965. This is an expansion of the set theory as a function of the character attributes of a member of the values instead of its values from 0 to 1. Thus the soft transformation of sets and representation brings about the difference in linguistic rules, which results in step-by-step feedback and expertise in development. The fundamental structure of the fuzzy system is shown in Fig. 1. There are four steps in the constant fuzzy logic, which are the intermediate logical quotient, the rule database of the site, the inference machine and the stutter (i.e., fuzzification, rule base, inference engine, and defuzzification).

Table 2
Mean and standard deviation of soil various parameters.

Characteristics	Category	Mean	Standard deviation
pH	Soil and water parameter	7.41	0.89
Electrical conductivity	Soil and water parameter	0.22	0.16
Organic matter	Soil parameter	1.10	0.35
Nitrogen	Primary nutrient	24.10	9.40
Phosphorus	Primary nutrient	30.96	15.52
Potassium	Primary nutrient	164.16	111.03
Calcium	Secondary nutrient	495.46	233.52
Magnesium	Secondary nutrient	22.38	17.27
Sulfur	Secondary nutrient	179.38	120.14
Sodium	Micro-Nutrient	1.66	1.07
Zinc	Micro-Nutrient	508.17	31.02
Manganese	Micro-Nutrient	33.37	45.51
Boron	Micro Nutrient	2.28	0.92
Molybdenum	Micro Nutrient	0.88	0.14
Copper	Micro-Nutrient	13.36	6.10

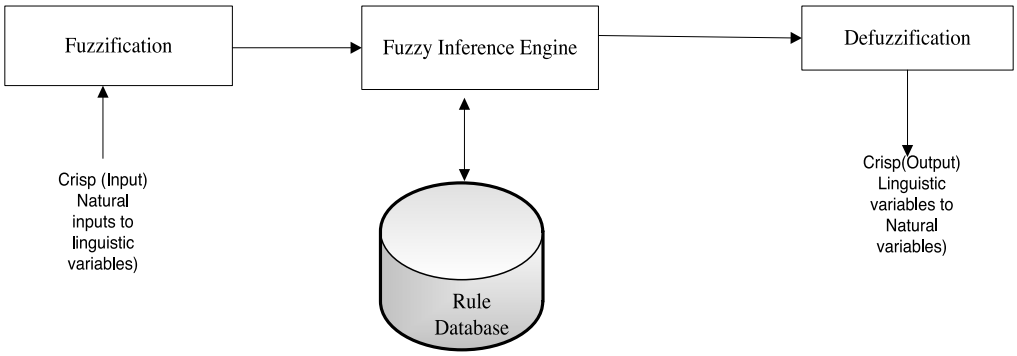

Fig. 1. Basic fuzzy system.

2.1. Support vector machine

Support Vector Machine is the most predominant data classification and regression techniques. The main goal of the support vector machine is to invent a hyper plane that can be used to increase the geometric margin limits of their classification error when given in two classes of linear differentiable problems. SVM is the most valuable system which is majorly concentrated to optimize structural risk assessment of real-time data. Generally, the support vector machine automatically assumes the accuracy of the machine that uses the machine itself and using its machine learning capabilities to increase computation. It also holds the position that matches the data. Kernel functions are initiated in the support vector machine classification, with a specific end goal of eliminating non-direct ones. The two main non-critical phases of the support vector machine in practice are the training stage and the testing stage. In an arrangement for generating examples, each has two class stages, one is the SVM training algorithm, and the other is a model that classifies new events.

2.2. Kernel selection

This section focuses on kernel selection functions. The SVM has qualities that only work within the chosen directions and the defined end boundaries. It operates within such confined limits; it results in an inappropriate result in the output that depends on real-time complex constants. Then, the kernel selection process is conducted and the final data set selection and parameters are determined, which depends on the complexity of the variables. The linear separable platform is fully integrated into the site, as the linear support of the data handling functions is handled by the SVM efficiently. Two packages of linear separable consists of two divisions, i.e., two separate

classes, first, the actual value is from the input vector data in the n-dimensional, and second is the data of class entries. Besides, it is hard on the edges since it has values based on real-time; hence the actual values of a hard margin can only be handled with a linear support vector machine. Simultaneously, linear kernel functionality alone is not sufficient for real-life problems, and the need for better assumptions on such direct real-time scenario functions creates a compulsory standard for testing other kernel systems. It is therefore forced to accept other kernel functions such as radial bias function, polynomial kernel function, and sigmoid function. Therefore among the four different kernel functions, the RBF has been deeply involved and has provided excellent accuracy. But, at the same time other kernel functions are also taken into account. Hence, the kernel mechanism of SVM is reviewed as follows. The four most commonly used kernel functions and parameters are listed in Table 2.

2.3. Fuzzy support vector machine (FSVM)

Even in this state also, the radial bias function alone does not help achieve high precision. This is because the nutritional value of the plant in this proposed system is considered based on the individual cascaded system. Hence, determining the quantity of fertilizer recommendation is more complicated, including a variety of systematic practices. In addition, to calculate the land suitability index is vital to get a positive yield on the ground. Such a process of land compatibility has been reviewed by [33]. So, in this case, the fuzzy combined support vector machine functions have been followed [7,28]. Meanwhile, due to the linear hard margin and plane gradient constants with a high dimensional feature, a slack variable is introduced. Assessment error indicates gradient variables such as maximum and minimum quantity, and the class-based linear optimization problem is the issue of all-in-one data points and the fuzzy membership values that imply the bulk of imported and reflected errors [14]. Therefore the concept of inner and outer boundaries allows using the support vector machine in fuzzy logic. On that account, to facilitate such a feature, the fuzzy combined support vector machine is used. Because fuzzy structure and support vector engine models are exposed openly by their excellent features, as a result, it has a high computational performance over other techniques. Therefore, the FSVM feature is intended to detect the classification hyperplane (x) divides data points. Classification is represented by a hyperplane. SVM has separate data based on the hyperplanes used for two-class issues. Given the training sample label pairs (A_iB_i) $i = 1\ldots.$ $A_i \in Q^n$, $B_i \in$ between interval $\{-1$ to $1\}$. Here, the SVM is determined by the intent of the highest hyperplane work and is marked by an orthogonal vector (w) and a function bias (b), respectively.

$$f(x) = w.\ x + b = 0$$

Even after the use of the vector engine, the n-dimensional hyperplane has a slightly misleading error in the linear segments of the hard edges, where the gradient variables are emitted from the existing errors and the deployed support vector engine mathematical equation is,

$$\xi_i(B_i = 1)A_i.w + b \geq +1 - \xi_i$$

$$\xi_i(B_i = 1)A_i.w + b \leq -1 - \xi_i$$

The combination of two slack variable functions along with SVM is as follows

$$\min \frac{1}{2} \left\| W^2 \right\| + C \sum_{i=1}^{L} \xi_i$$

The constant C represents the error classification data functions which are based on a constant value and value violation variables. The standard value constants and value violations are ready to pay the misclassified functions. Therefore, data points associated with high vague membership values are essential. At the same time, small vague membership values, the effect of data points related to the excessive classification of the hyperplane are small. In the meantime, it seems to be an error rating as a misclassification data point at another view. As a result, it is possible to find a stronger hyperplane overloaded by allowing some misleading classifications to increase FSVM limits and

lower the key data points. For this purpose, the Lagrange related optimization problem has been built [14].

$$Max \sum_{a-1}^{f} Z_i - \frac{1}{2} \sum_{a=1}^{f} \sum_{b=1}^{f} Z_a Z_b \delta_a \delta_b K(\eta_a, \eta_b)$$

$$\text{Subject to } \sum_{a=1}^{f} Z_1 \delta_1 = 0$$

$$0 \leq Z_a \leq C, \text{a} = 1, 2, 3, \ldots \text{f}$$

Z_a is the Lagrange multiplier and $K(\eta_a, \eta_b)$ is the inner product of the feature vectors of η_a, η_b in the feature space like,

$$K(\eta_a, \eta_b) = \mu(\eta_a)^T \mu(\eta_b)$$

where $K(\eta_a, \eta_b) = \mu(\eta_a)^T \mu(\eta_b)$ is the kernel function and it implicitly defines the features of μ. Due to the difference between the dual optimization of the original support vector machine and the dual optimization of the fuzzy support vector machine, the karush–kuhn tucker theorem has been applied. After applying the karush–kuhn tucker theorem, the classification of hyper plane is represented by,

$$\sum_{a=1}^{f} Z_a \delta_a K(\eta_a, \eta) + S = 0$$

Then the final decision function of FSVM function is,

$$h(x) = \text{sgn}\left(\sum_{a=1}^{f} Z_a \delta_a K(\eta_a, \eta) + S\right)$$

Typically, designing any forecast model has to consider various parameters associated with its predefined demands because that depends on the design of an application. Therefore, the parameters of the relevant design application are determined by the support vector machine for requests. It is collected by field experts and has a real domain profile that evaluates the complexity of the priority variables to detect previously available output using various charts. Therefore, it gives relevant results when using a vector support machine combined with fuzzy logic to handle complex constants within finite limits. For instance, a part of this research has three different seasonal conditions like Rabi season, Kharif season, and summer season. Each has a linear divergence of exercise because the seasonal factors are changing over time. Therefore, the linear fuzzy membership functions of the seasonal constants are as follows [3,27].

$$\theta_1 = \partial(\omega_S)$$

Therefore the fuzzy membership functions in this study have been used as given below.

$$\theta_1 = (1 - \delta)\left(\frac{\omega_S - \omega_1}{\omega_P - \omega_1}\right)^i + \delta$$

where 'i' is the dimension of fuzzy membership functions, which are used in the range of i values for kernel functions such as linear, polynomial, radial bias functions and sigmoid.

2.4. Throughput prediction using support vector machine and fuzzy inference model

This part focuses on the design of an agriculture yield prediction DSS, which comprises an intelligent agent. If the same size with regard to the prediction of production is required; it is necessary to make fertilizer applications and proper fixing. The goal of the research is to design the system that depends on the needs of farmers. At the same time, determining the quantity of fertilizer recommendation has some difficulties, which indeed leads to a variety of systematic practices. Apart from that, to calculate the cropping pattern of the land suitability index is vital to get a positive yield on the ground. Here, such a system of land compatibility is derived from this research report [33]. Not only that but also our expert recommendations have mentioned the same. Therefore to facilitate

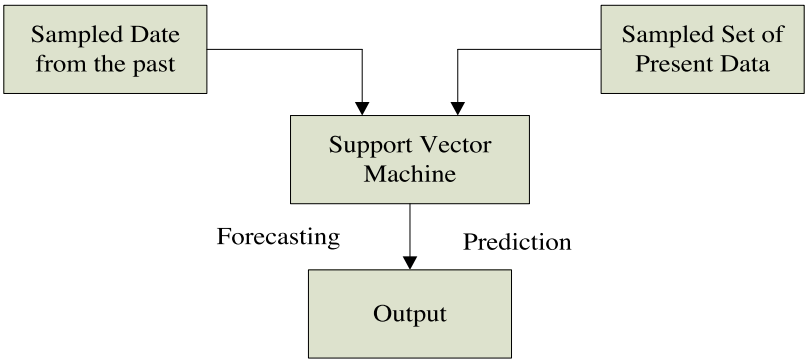

Fig. 2. Productivity prediction model using SVM.

such a feature, the fuzzy logic and support vector machine is used. Because fuzzy structure and support vector engine models are exposed plainly by their excellent features, as a result, it has a high computational performance over other techniques. Fig. 2 shows the predicted prediction model using a support vector machine.

2.5. Modeling for an FSVM decision support system in fertilizer balance

The methodology of its adaptation has been used in this way. It has been followed by a graphical approach to satisfy the fertility through the World Food and Agriculture analysis work [47]. The agricultural process consists of various factors such as rain, temperature, humidity, and three different types of fertilizers. Minimum law factors, such as climate variables and primary nutrient units, always had a hand and the productivity of several other factors. Nevertheless, climate models of the minimum law is unchangeable, while others are changed variables; it is also combined with three nutrient types. Our paper refers to its completions with mathematical expressions and reveals the unique configuration structure definition based on a soil structure of land [44].

$$E_x = \frac{\left(\int_{\Delta_1}^{\mu} X * \int_{\Delta_2}^{\mu} Y * \int_{\Delta_2}^{\mu} Z\right)}{(IndispensableValueX * IndispensableValueY * IndispensableValueZ)} \tag{1}$$

In this Eq. (1) X, Y, and Z are the primary, secondary, and micronutrient, respectively. According to the expert's recommendations, the value of μ is always at 5, since it is based on the Food and Agriculture Organizations announcements. The value of $\Delta 1$, $\Delta 2$, and $\Delta 3$ is derived from the actual values of nutrients strength in the land. The nutrients required for a particular crop is determined from the expert suggestion. These two form the lower limit and upper limit of the integration procedure adopted in calculating the recommendations for a particular crop. Access to crop-concentrated land may not be a negative Ex value because it will be available to a specific crop of existing nutrient level limits. For instance, the value of Ex $= 0$, as it has been pronounced, does not require any recommendations to the soil, except when the value of Ex $= 1$ requires 100% recommendations. If the value of Ex < 0.50 and Ex > 0.50, the recommendations are 75% and 25%, respectively. Therefore, it is best to get the crop based on the recommendations of experts through the DSS, and depending on the above factors a basic rule of such a system is designed and implemented. The need for designing an embedded decision-making system does the crop selection and increases its productivity depending on the number of minerals in the soil. As the minerals contained in the soil vary from one land to another, therefore, each nutrient concentration in the proposed system is considered to be the individual benefit of a cascaded system. Hence, the individual concentration levels are multiplied by the collective advantages of the stratosphere. The soil samples are distributed throughout the land to deal with the nutrient density of soil through soil testing. Each particular nutritional concentration shows a small variation. Therefore, with the concepts of conjecture, it is inserted into a stable loop of the fuzzy logic system. Generally, agriculture is a board that includes many inputs and outputs, so it has to deal with many problems when it is converted to an organized system because all of the parameters in each of the subhead in agriculture have subsided and dependent on one another. Nevertheless, it has been modified into a register transfer level design, which can quickly get outputs and is then implemented in an FPGA.

2.6. Predictive modeling

In the field of assessing the needs of fertilizer, fuzzy logic is used with different approaches. Similarly, to determine the fertilizer scale in agriculture, the support vector machine is also used with other approaches. The fuzzy logic and support vector machines in determining the inputs, evaluating the product and designing the applications on these applications are the main reasons for the use of the approach and the ability to coordinate with the expert knowledge. In addition, they are stable and more important in the implementation of terminal expressions, mainly through linguistic exposures. It is an indistinct and inherently complex concept to define the difficulties or measures and also, it is easier to create an exhibiting structure such as modeling logic structure and support vector machine models due to the increasing interest towards the contribution approaches in fertilizer recommendations. At the end of the vector mechanical model development process, the user can serve as an enhanced adherence and a better technique with the fuzzy logic in such an attempt. The database has already been stored on land profile details. Meanwhile, production is expected to be based on the expert rule and predicted production level. Based on the predefined data, the instrument can give a level of crop yield, the initial symptom of their condition and the percentage of recommendations. Agricultural field experts are the first entity to provide rankings to measure the required health parameters of crop influence factors. The professional domain experts are the second entity who interact with the intelligent system, the third entity confirm or deny the measurement efficiency produced by an intelligent system. The Intelligent Embedded DSS validates all crop influenced factors and provides the stepwise procedure to utilize the fertilizer in terms of percentage for improving the crop productivity. For this study, the study area has been monitored from 2013 to 2015. The idea of FSVM decision support systems has served as a very useful organizing principle, although with a solid foundation for maintaining soil fertility. Consequently, the final form of the system is an intelligent embedded decision support system agent which helps to predict crop yields through various crop influence factors. The smart device plays a role in finding a yield based on crop influence factors in a regular crop season and uses FSVM to find future crop yield conditions. Based on the predefined data, the instrument can give a level of crop yield, the initial symptom of their condition and the percentage of recommendations. Early data provided by the Agricultural Department experts are stored in a profile with crop influence factors. The use of the yield knowledge base is subject to the diagnostic process of data crops. The recommendation process is based on the data of crop influence factors. The hypothetical machine predicts the future yield—status of crop influence factors based on fuzzified data. Based on the computation, the smart system gives an identity about the potential production of crop yields. Fuzzy membership functions have been determined by the British Association Meeting 1955, Food and Agriculture organizations, and the team of experts in our field. This system is designed based on our analysis of the field and the questions presented to the beneficiaries. At the beginning of this study, we conducted a questionnaire survey; by this, we got a brief overview of some explanations in which we realized that some of these should be installed. Besides, the system is established as a result of increasing productivity, reducing costs and ensuring the quantum of future productivity. Initially, with the full cooperation of the farmers, the soil sample is collected and all the essential constants of crop cultivation parameters are examined as per their opinion, and the future crop production quantity is predicted. Similarly, the water sampling is collected and taken into consideration for all the acidic, alkaline conditions and the optimal processes need for irrigation is ensured. On the other hand, the proper season for the crop is kept within the database. Through this, future yields from the true position of the parameters are calculated, and if any of the defects arise, it is fixed according to the recommendations of the field experts. Since agriculture is dependent on a wide range of constants and parameters, the challenges in establishing the decision-making system are very high. As a result of the impact, all the agricultural parameters are transformed into FSVM and the first step is to designate a system with the recommendations of the field experts, and then designate as an embedded system.

3. System design

It consists of more than one and individual blocks of crop influenced factors. It makes farmers to cultivate crop along with favorable yields properly. The database will already have localized and terrestrial system data, such as the dynamite line, linear elements, and the factors that can participate in the operating effects of the plants. Here, the intelligent embedded DSS is basically to represent the FSVM. A database of profiles about agricultural lands is used; at least three entities are required to operate the intelligent embedded agriculture DSS. The soil is the first entity, which includes primary, secondary and micronutrients. The remaining factors that are involved in the growth

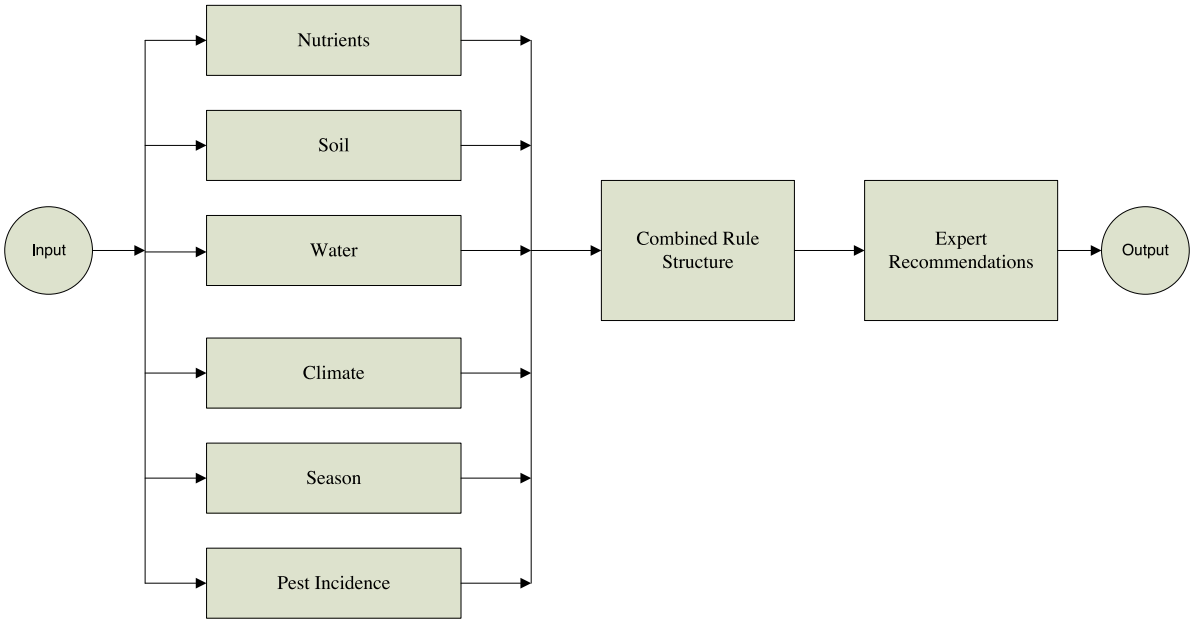

Fig. 3. Is explained as a general trend of expert recommendations with crop influence factors to boost production.

of the crop are the second entity. First and second entity interconnects the DSS to ensure the minimum throughput required. Finally, the intelligent decision support system is viewed as a third entity because this is only to predict the future yield in advance; it also suggests the scale of recommendation when the land meets the targeted output of farmers. Fig. 3 illustrates the system to boost the production of the crop. Here, the system is broadly divided into six subcategories.

3.1. Design of FSVM simulation method

The available open-source library file is used to build the FSVM model. Initially, all data sets are converted to the FSVM training system using the random selection technique. For instance, the data sets are transformed as follows. The types of product variants are normalized from -1 to 1, for example, low, moderate, good, and excellent, respectively. Besides the different types of kernel functions, such as polynomial, linear, sigmoid, and radial bias functions are also used for simulation.

3.2. Xfuzzy environment

Xfuzzy is a platform that automatically generates the register transfer level design [11]. The Xfuzzy tool is designed for data entry functionality and data member functionality for input recording evaluation units provided by a memory, gate, and NOR, EX-NOR, or XOR in addition to many pipe amplifiers, counters, counters, and sub force. Performance in the complete sequence of the aforementioned continuity has been successful. The Xfuzzy VHDL generic system generator generates the VHDL code when the code for the Xilinx platform is uploaded and the technical package parameters are listed in Table 3. The recommendations of the field experts are first converted into a rule-based system with appropriate membership functions, and then the RTL of the design is created automatically [12,36]. Results of total engagements are discussed in the following chapters. To get the effect of better output hypothesis, Xfuzzy platform is used. The center of the most obscure mean and gravity can be used for steering. The reason behind COG and FM redirection selection is to fulfill the expert recommendations of the domain and to provide reasonably clear release values. Zynq micro-board specifications used the IP block design for the implementation of the designed process, and then the specified parameters are adequately organized.

Table 3
Kernel function and its parameters.

Kernel function	Expression	Parameter
Linear kernel function	$A(z, z_i) = (z.z_i)$	
Polynomial kernel function	$A(z, z_i) = (x(z.z_i) + y)^b$	x, y, b
Radial bias function	$A(z, z_i) = \exp\left\{-\frac{\lvert z-z_i\rvert^2}{2c^2}\right\} = \exp(-\delta \left\lvert z - z_i{}^2\right\rvert)$	c, δ
Sigmoid function	$A(z, z_i) = \tanh(x(z.z_i) + y)$	x, y

Table 4
The resource utilization of the intelligent embedded DSS.

Logical resource	Used
Slice logic utilization	
Number of occupied logic cells	11
Number of slice registers	15
Number of lookup tables	22
Slice logic distribution	
Number with unused Flip	12
Number with unused LUT	1
Specific features utilization	
Total registers	27
Number of bonded IOBs	11
Average fan out	2.83
Minimum period	2.132 ns
(Maximum Frequency : 469.120 MHz)	
Minimum input arrival before clock: 0.608 ns	
Maximum output register time after clock: 1.108 ns	
Maximum combinational path delay: 0.792 ns	

3.3. Implementation of FPGA based DSS

The intelligent DSS based embedded system is implemented in the FPGA Zynq family xc7z010-2clg400 target device. In FPGA implementation, the Xfuzzy platform is used. Xfuzzy is the platform it creates the automatic RTL netlist for a given specific fuzzy logic system [10]. Therefore Xfuzzy is used to generate the corresponding bitstream pattern for the overall system and further it is synthesized. The details of downloading the bitstream pattern of field-programmable gate array are derived in [48]. The SRAM is built in the FPGA, which helps to store configuration data and structured information. The values of these parameters are calculated using the values of its associated member functions. Here it represents the functions of the FSVM membership functions. Different possibilities based on the functioning values of these membership functions are poor, moderate, good, and excellent, which is already described in previous sections. The future description of the throughput is reported, depending on the conditions in these seasons, which are the most likely of these four types of possibilities. Fuzzy based decision-making in the hardware occupies excessive logical evidence when establishing a support system. It is, therefore, possible to reduce the efficiency of the designing system and reduce the speed, but occupying the general logic resource utilization of Xfuzzy is completely avoided. The designed agricultural embedded system schematic is shown in Fig. 4. Table 4 illustrates the resource utilization of DSS.

4. Results and discussions

By all means, we use a fuzzy support vector machine to build the correct solution that identifies the agricultural productivity from the questionnaires which are collected from the farmers through the field works.

The system is tested for 150 farmlands. In this actual experiment, the primary nutrient, secondary nutrient, micronutrients, and other factors of seasonal conditions such as temperature, humidity, rainfall, water and soil condition and environmental impacts such as plant diseases are taken into account. Initially, the soil nutrients reports of the land are given. Based on the soil nutrients reports, the throughput of the crop and needed fertilizer levels are

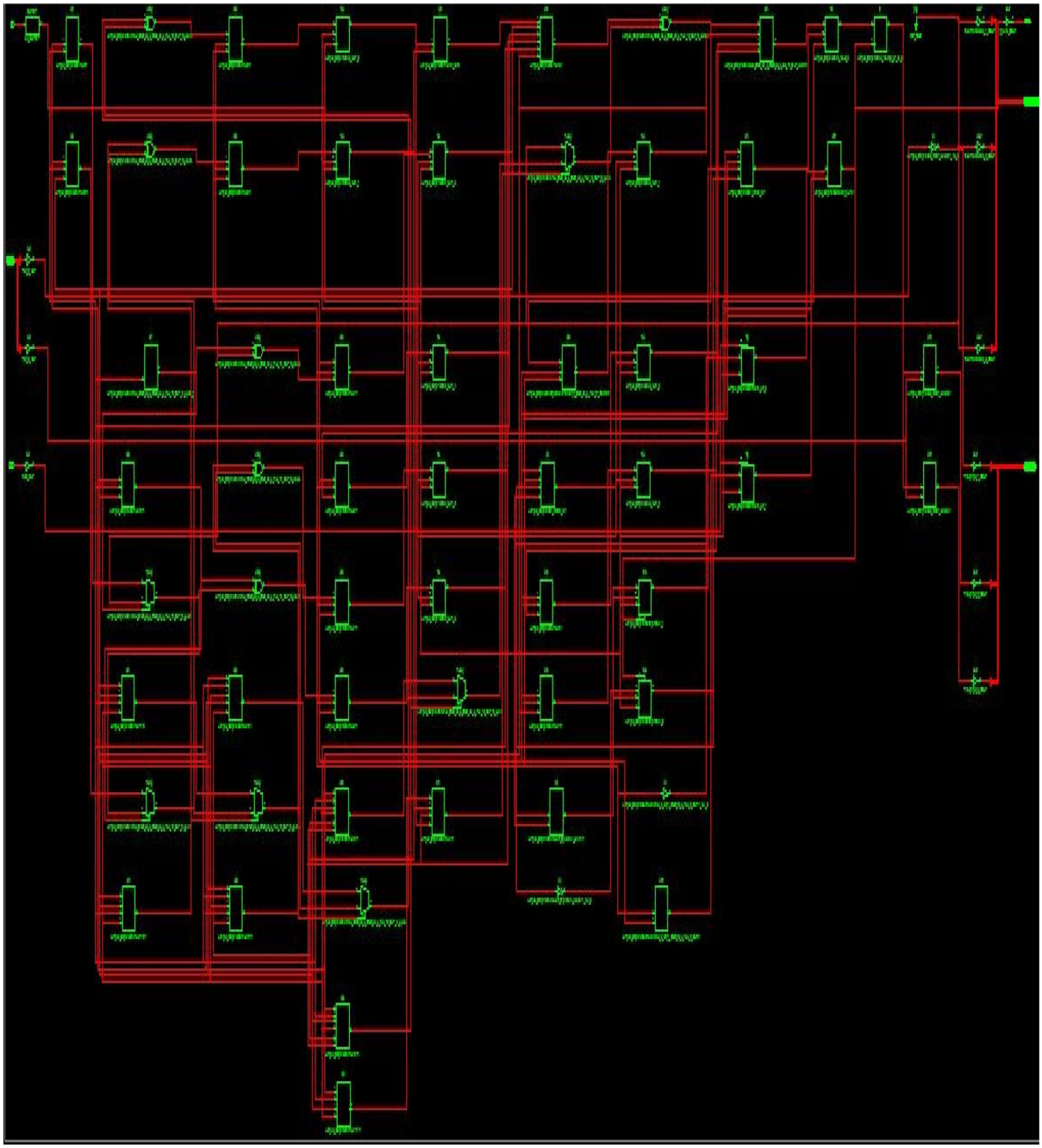

Fig. 4. Embedded decision support system schematic structure in Xilinx platform.

computed by the system. Using the parameter, the corresponding membership functions are computed and depending upon the availability of soil nutrients the system may suggest the exact nutrients need. The essence of this creation, however, is that it helps in figuring out the yield and production that can be achieved in soil conditions and favorable seasoning. For instance, humidity, temperature, and rainfall can generate good yields at average levels. Similarly, whether it is micronutrient or nutrient, the own scale of parts per million (ppm) are untouched or when it exceeds, it causes visual defects to the plant for example as far as boron is concerned, it does not affect the plant when it is 0–5ppm scale. Based on these criteria, the decision support system is set up and tested in farmers' fields and

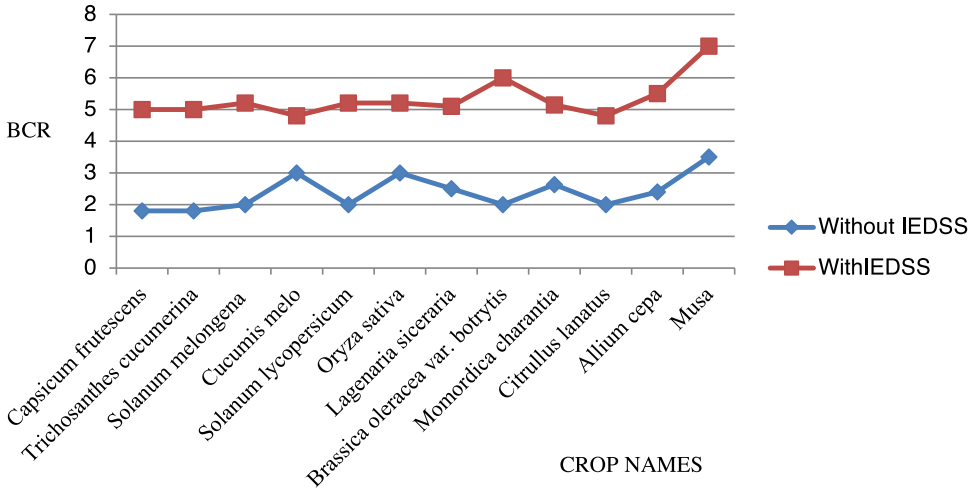

Fig. 5. Comparison chart of BC ratio.

Table 5
Kernel functions and its accuracy.

Kernel types	Linear kernel	Polynomial kernel	Radial bias function	Sigmoid
Accuracy	64.29	95.38	98.18	72.73

productivity can be achieved if the other factors cooperate. Table 2 describes the sample soil analysis report, which is used in the proposed machine to predict future production of the Rabi season. We have provided three-yearly recommendations to the farmers to cultivate 12 crops with the intellectual information obtained through this format. It has a total of 12 seasons with three seasons per year as per the Indian crop cultivation period. Based on these three years, the rate of three seasons per year is tested at the field level and the graph is designed and cataloged. The plotted graph represents the average BC ratio for three years, with the rate of 3 seasons per annum; in addition, it shows the influence of the decision support system on the benefit–cost ratio with and without the usage of DSS. Fig. 5 graphical representation shows, the BC ratio is higher when the farming community follows the DSS and the benefit–cost is less when the support system does not follow. In some cases, the production has been greatly affected by tragic circumstances. The annual production of the northeast region decreased and their income hampered due to the heavy rainfall in 2015. The proposed system has come up as an endorsement of the DSS made by incorporating such real facts. This decision has been made with the notion that all other non-parameters other than parameters that cause plant crop impact are involved in the development of the crop and helps to increase the yield. The proposed research work is to target only the design of the installed system, which did not take into account the speed of the computer and its technical specifications, because it plans to do this as a future comprehensive release. Among the crops, the 12thcrop Musa has a high yield rate of three years, which will receive ground level after introducing the overall system-based design.

The FSVM based system found a better throughput level of classification performance with better accuracy. Accordingly, the accuracy of the linear kernel, polynomial kernel, radial bias function, and the sigmoid are given in Table 5. Overall, the radial bias function has been provided the highest level of accuracy.

The results of the research have indeed confirmed our predictions. Implementing through a system of fuzzy logic and support vector machine, the profitability ratio has proven to be high, and our farmers have been sharing their heartfelt success stories with the idea that the benefit–cost ratio has risen after adopting these farming practices.

4.1. System comparative analysis

While considering the literature review, more studies have predicted the future crop yield without using the fuzzy or limited parameters, and only a few used decision support system to predict the crop yield and with less number

Table 6
Comparison of the methods.

System	Recommendations type	System type crop yield prediction	Prediction report/ accuracy
Crop selection Using expert system [23]	Recommendations not provided	Fuzzy based expert system	Not reported
FDSSICF [44]	Recommendations provided	Decision support system	Reported/not applicable
Proposed system	Recommendations provided	FSVM used FPGA based System	Reported/95 (BCR – 4.5/5)

of parameters. The other is that previous works are not checked at the field level. Therefore, we have to come to know that this proposed system cannot be compared with any prior research works. Meanwhile, in this research, the productivity of agriculture is calculated and tested at field level. The most important factor of the proposed system is not only to predict the crop productivity based on crop selection but also for the production capacity of the crop. On the other hand, one of the features of this proposed system is the ability to increase productivity in order to respect the interests of the farmer. In the interest of the peasants concerned, it is first confirmed that the level of attainment of the minimum requirement is achieved. Most of the effort is only related to the success of their traditional character. Prior to that, such research has already been done in India's behaviors in which the method of crop selection is used only by the use of fuzzy logic-based expertise, except that the recommendations of the experts are not given to the farmers. Table 6 describes the comparison between the methods.

5. Conclusions

This paper describes the development of an FPGA based intelligent embedded decision support system for agriculture applications. Presently, Agriculture is a decentralized or noncore phenomenon system even in a country like India. Even if it is considered to be a far-reaching farmer by this uninterrupted state, it will affect the gross domestic product in the long term when it crosses a particular position. Consequently, algorithms on existing international practices have been adopted to transform agricultural variables as soft computing systems. This DSS calculates and predicts the advance yield of the agricultural crop on the basis of the plant's growth influence factor and various plant parameter multiplication constants.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

The authors wish to acknowledge the financial supporters, Government of India, Ministry of Agriculture and Farmer's welfare for their work provided through the scheme of National Agriculture Development Programme through Government of Tamil Nadu and supported by Department of Horticulture and Plantation Crops.

References

[1] Fuzzy logic design tools, institute of microelectronic de sevilla (imse-cnm).
[2] Trace Elements and Plant Growth- Nature- 621- 623- 176 44831476-4687 http://dx.doi.org/10.1038/176621a0 http://dx.doi.org/10.1038/176621a0ref1.
[3] The Classification of Electrocardiograph Arrhythmia Patterns using Fuzzy Support Vector Machines.
[4] J.H. Agarwal, Electronic aids for enhancement of agricultural production, IETE Tech. Rev. 10 (1) (1993) 57–64.
[5] R. Ali, A.K. Pal, S. Kumari, M. Karuppiah, M. Conti, A secure user authentication and key-agreement scheme using wireless sensor networks for agriculture monitoring, Future Gener. Comput. Syst. (2017).
[6] Mohammad J. Anar, Zhulu Lin, Gerrit Hoogenboom, Vakhtang Shelia, William D. Batchelor, Jasper M. Teboh, Michael Ostlie, Blaine G. Schatz, Mohamed Khan, Modeling growth, development and yield of Sugarbeet using DSSAT, Agric. Syst. 169 (2019) 58–70.
[7] A.M. Andrew, An Introduction to Support Vector Machines and Other Kernel-Based Learning Methods By Nello Christianini and John Shawe-Taylor, Cambridge University Press, Cambridge, ISBN: 0-521-78019-5, 2000, p. xiii+ 189, 2000, Robotica, 18(6), 687-689.

[8] A. Barriga, I. Baturone, S. Sánchez, R. Senhadji, Hardware synthesis of fuzzy systems using the Xfuzzy development environment, 1970.
[9] Angel Barriga, Santiago Sánchez-Solano, I. Baturone, D.R. López, Francisco José Moreno Velo, Federico Montesino Pouzols, Nashaat M. Hussein, Piedad Brox, A. Gersnoviez, New features of the fuzzy logic development environment Xfuzzy, 2006.
[10] A. Barriga, Santiago Sánchez-Solano, P. Brox, Alejandro Cabrera, Iluminada Baturone, Modelling and implementation of fuzzy systems based on VHDL, Internat. J. Approx. Reason. 41 (2) (2006) 164–178.
[11] Barriga Barrios, Santiago Sánchez Solano Ángel, Carlos J. Jiménez, D. Galán, Diego R. López, Automatic synthesis of fuzzy logic controlers, Mathware Soft Comput. 3 (3) (1996) 1996.
[12] María Brox, Santiago Sánchez-Solano, Ernesto del Toro, Piedad Brox, Francisco Jose Moreno-Velo, CAD tools for hardware implementation of embedded fuzzy systems on FPGAs, IEEE Trans. Ind. Inf. 9 (3) (2013) 1635–1644.
[13] M. Brox, S. Sanchez-Solano, L. Delgado, XFVHDL4: A hardware synthesis tool for fuzzy systems, in: 2011 11th International Conference on Intelligent Systems Design and Applications, 2011, pp. 385–390.
[14] Aslı Çelikyılmaz, I. Burhan Türkşen, Fuzzy functions with support vector machines, Inform. Sci. 177 (23) (2007) 5163–5177.
[15] J. Chen, M.J. Patyra, VHDL modeling of a multivariable fuzzy logic controller hardware system, in: Fuzzy Systems, 1994. IEEE World Congress on Computational Intelligence, in: Proceedings of the Third IEEE Conference on, IEEE, 1994, pp. 129–132.
[16] Xiaoping Chen, Zhiming Qi, Dongwei Gui, Matthew W. Sima, Fanjiang Zeng, Lanhai Li, Xiangyi Li, Zhe Gu, Evaluation of a new irrigation decision support system in improving cotton yield and water productivity in an arid climate, Agricult. Water Manag. 234 (2020) 106139.
[17] Jung-Hsien Chiang, Pei-Yi Hao, Support vector learning mechanism for fuzzy rule-based modeling: a new approach, IEEE Trans. Fuzzy Syst. 12 (1) (2004) 1–12.
[18] Yu Duan, Hui-Yan Zhang, Xue-Bo Jin, Ji-Ping Xu, Ting-Li Su, Evaluation model of agricultural irrigation water quality based on projection pursuit and fuzzy support vector, in: 2017 Chinese Automation Congress (CAC), IEEE, 2017, pp. 3786–3790.
[19] Maryam Farhadian, Parisa Shokouhi, Parviz Torkzaban, A decision support system based on support vector machine for diagnosis of periodontal disease, BMC Res. Notes 13 (1) (2020) 1–6.
[20] Kit Fai Fung, Yuk Feng Huang, Chai Hoon Koo, Majid Mirzaei, Improved SVR machine learning models for agricultural drought prediction at downstream of Langat River Basin, Malaysia, J. Water Clim. Change (2019).
[21] Bhaskar B. Gardas, Rakesh D. Raut, Naoufel Cheikhrouhou, Balkrishna E. Narkhede, A hybrid decision support system for analyzing challenges of the agricultural supply chain, Sustain. Prod. Consum. 18 (2019) 19–32.
[22] José Miguel Guerrero, María Guijarro, Martin Montalvo, Juan Romeo, Luis Emmi, Angela Ribeiro, Gonzalo Pajares, Automatic expert system based on images for accuracy crop row detection in maize fields, Expert Syst. Appl. 40 (2) (2013) 656–664.
[23] A. Kapoor, A.K. Verma, Crop selection using fuzzy logic-based expert system, in: Applications of Soft Computing for the Web, Springer, Singapore, 2017, pp. 121–137.
[24] Rakesh Kaundal, Amar S. Kapoor, Gajendra P.S. Raghava, Machine learning techniques in disease forecasting: a case study on rice blast prediction, BMC Bioinform. 7 (1) (2006) 485.
[25] Barnabas Kiiza, Glenn Pederson, ICT-based market information and adoption of agricultural seed technologies: Insights from uganda, Telecommun. Policy 36 (4) (2012) 253–259.
[26] J. Kumar, N. Bhardwaj, FPGA based advanced sowing and planting equipment controller design, in: Emerging Trends in Electronic and Photonic Devices & Systems, 2009. ELECTRO'09. International Conference on, IEEE, 2009, pp. 120–123.
[27] S. Lee, E. Paik, Y. Lee, Time series data pattern classification using fuzzy membership functions and support vector machines : KOSPI 200 : Korea composite stock price index 200, Artificial Intelligence 103 (305) (2003) 43–45.
[28] C.F. Lin, S.D. Wang, Fuzzy support vector machines, IEEE Trans. Neural Netw. 13 (2) (2002) 464–471.
[29] Holger R. Maier, Graeme Clyde Dandy, Modelling cyanobacteria (blue–green algae) in the River Murray using artificial neural networks, Math. Comput. Simulation 43 (3–6) (1997) 377–386.
[30] Julio Mar-Ortiz, Norberto Castillo-García, María D. Gracia, A decision support system for a capacity management problem at a container terminal, Int. J. Prod. Econ. 222 (2020) 107502.
[31] R.L. McCown, Changing systems for supporting farmers' decisions: problems, paradigms, and prospects, Agric. Syst. 74 (1) (2002) 179–220.
[32] Joshua Mendoza-Jasso, Gerardo Ornelas-Vargas, Rodrigo Castañeda Miranda, Eusebio Ventura-Ramos, Alfredo Zepeda-Garrido, Gilberto Herrera-Ruiz, FPGA-based real-time remote monitoring system, Comput. Electron. Agric. 49 (2) (2015) 272–285.
[33] Mohsen B. Mesgaran, Kaveh Madani, Hossein Hashemi, Pooya Azadi, Iran's land suitability for agriculture, Sci. Rep. 7 (1) (2017) 7670.
[34] Jesus Roberto Millan-Almaraz, Rene de Jesus Romero Troncoso, Ramon Gerardo Guevara-Gonzalez, Luis Miguel Contreras-Medina, Roberto Valentin Carrillo-Serrano, Roque Alfredo Osornio-Rios, Carlos Duarte-Galvan, Miguel Angel Rios-Alcaraz, Irineo Torres-Pacheco, FPGA-based fused smart sensor for real-time plant-transpiration dynamic estimation, Sensors 10 (9) (2010) 8316–8331.
[35] Eric Monmasson, Marcian N. Cirstea, Cirstea FPGA design methodology for industrial control systems—A review, IEEE Trans. Ind. Electron. 54 (4) (2017) 1824–1842.
[36] Francisco José Moreno-Velo, Iluminada Baturone, Angel Barriga, Santiago Sánchez-Solano, Automatic tuning of complex fuzzy systems with Xfuzzy, Fuzzy Sets and Systems 158 (18) (2007) 2026–2038.
[37] Satoshi Nakano, Ayu Washizu, Induced effects of smart food/agri-systems in Japan: Towards a structural analysis of information technology, Telecommun. Policy 42 (10) (2018) 824–835.
[38] R. Natarajan, J. Subramanian, E.I. Papageorgiou, Hybrid learning of fuzzy cognitive maps for sugarcane yield classification, Comput. Electron. Agric. 127 (2016) 147–157.

[39] P. Pandey, R. Litoriya, A predictive fuzzy expert system for crop disease diagnostic and decision support, in: Fuzzy Expert Systems and Applications in Agricultural Diagnosis, IGI Global, 2020, pp. 175–194.
[40] S.N. Pandey, A.K. Mukherjee, A.K. Jain, Prospects of electronics in agriculture at post harvest stages in commercial crops, IETE Tech. Rev. 8 (6) (1991) 351–354.
[41] Elpiniki I. Papageorgiou, Athanasios T. Markinos, Theofanis A. Gemtos, Fuzzy cognitive map based approach for predicting yield in cotton crop production as a basis for decision support system in precision agriculture application, Appl. Soft Comput. 11 (4) (2011) 3643–3657.
[42] Tiago Pinto, Tiago M. Sousa, Isabel Praça, Zita Vale, Hugo Morais, Support vector machines for decision support in electricity markets' strategic bidding, Neurocomputing 172 (2016) 438–445.
[43] G. Prabakaran, D. Vaithiyanathan, Madhavi Ganesan, A farmer-friendly initiative, in: Grassroots, A Journal of Press Institute of India, Research Institute for Newspaper Development, 2017.
[44] G. Prabakaran, D. Vaithiyanathan, Madhavi Ganesan, Fuzzy decision support system for improving the crop productivity and efficient use of fertilizers, Comput. Electron. Agric. 150 (2018) 88–97.
[45] Rashmi Priya, Dharavath Ramesh, Ekaansh Khosla, Crop prediction on the region belts of India: a Naïve Bayes MapReduce precision agricultural model, in: 2018 International Conference on Advances in Computing, Communications and Informatics (ICACCI), IEEE, 2018, pp. 99–104.
[46] M. Ray, D.C. Roy, A. Zaman, Evaluation of rice (Oryza sativa)-based cropping systems for increasing productivity, resource-use efficiency and energy productivity in coastal west bengal, Indian J. Agron. 61 (2) (2016) 131–137.
[47] R.N. Roy, R.V. Misra, J.P. Lesschen, E.M.A. Smaling, Assessment of Soil Nutrient Balance: Approaches and Methodologies, Food & Agriculture Organization, 2003, No. 14.
[48] Santiago Sánchez-Solano, María Brox, Ernesto del Toro, Piedad Brox, Iluminada Baturone, Model-based design methodology for rapid development of fuzzy controllers on FPGAs, IEEE Trans. Ind. Inf. 9 (3) (2013) 1361–1370.
[49] U. Segundo, J. López-Cuadrado, L. Aldamiz-Echevarria, T.A. Pérez, D. Buenestado, A. Iruetaguena, J.M. Pikatza, Automatic construction of fuzzy inference systems for computerized clinical guidelines and protocols, Appl. Soft Comput. 26 (2015) 257–269.
[50] Yize Sun, Shiqing Tang, Zhuo Meng, Yiman Zhao, Yunhu Yang, A scalable accuracy fuzzy logic controller on FPGA, Expert Syst. Appl. 42 (19) (2015) 6658–6673.
[51] Francisco José Moreno Velo, I. Baturone, Santiago Sánchez-Solano, Angel Barriga, Xfuzzy 3.0: A development environment for fuzzy systems, in: EUSFLAT Conf., 2001, pp. 93–96.
[52] F.J.M. Velo, L. Baturone, S.S. Solano, A. Barriga, Rapid design of fuzzy systems with Xfuzzy, in: The 12th IEEE International Conference on Fuzzy Systems, 2003. FUZZ '03., Vol. 1, 2003, pp. 342–347.
[53] Peter M. Vitousek, Rosamond Naylor, Timothy Crews, M.B. David, L.E. Drinkwater, E. Holland, P.J. Johnes, et al., Nutrient imbalances in agricultural development, Science 324 (5934) (2009) 1519–1520.
[54] L.-X. Wang, Jerry M. Mendel, Generating fuzzy rules by learning from examples, IEEE Trans. Syst. Man Cybern. 22 (6) (1992) 1414–1427.
[55] ZhiMin Yang, Jing Liang, GuangLi Liu, Application of strong fuzzy support vector machine on weather early warning of rice blast, J. China Agric. Univ. 15 (3) (2010) 122–128.
[56] Helin Yin, Yeong Hyeon Gu, Chang-Jin Park, Jong-Han Park, Seong Joon Yoo, Transfer learning-based search model for hot pepper diseases and pests, Agriculture 10 (10) (2020) 439.
[57] Yongli Zhang, Sanggyun Na, A novel agricultural commodity price forecasting model based on fuzzy information granulation and MEA-SVM model, Math. Probl. Eng. 2018 (2018).